

Figure 1: EPSP ratio (after/before) for various conditions. The plot shows data points with error bars (SEM) and fitted curves (χ^2 30.0, n 11) for the 'fitted' region (left of the dashed line) and validation data (χ^2 283.5, n 29) for the 'validation' region (right of the dashed line). The y-axis ranges from 0.50 to 2.25. The x-axis lists 30 conditions. A horizontal line at 1.00 indicates no change. A vertical dashed line at the 10th condition separates the fitted and validation regions.

Condition	EPSP ratio (after/before)	Region
10Hz_10ms	1.00	fitted
10Hz_-10ms	0.75	fitted
0.1Hz_dt+10ms	1.00	fitted
0.1Hz_dt-10ms	0.78	fitted
20Hz_dt+10ms	1.28	fitted
40Hz_dt+10ms	1.35	fitted
50Hz_dt+10ms	1.15	fitted
50Hz_dt+10ms NMDAR bl.	1.02	fitted
0.1Hz_dt-10ms mGluR bl.	1.00	fitted
S07 pair	1.65	fitted
S07 pair NO bl.	1.25	fitted
10Hz_5ms	1.05	validation
10Hz_-10ms mGluR bl.	1.02	validation
20Hz_dt-10ms mGluR bl.	1.05	validation
0.1Hz_dt-10ms NMDAR bl.	1.02	validation
20Hz_dt-10ms NMDAR bl.	1.02	validation
50Hz_dt+10ms mGluR bl.	1.02	validation
0.1Hz_dt-25ms	1.30	validation
0.1Hz_dt-120ms	0.70	validation
0.1Hz_dt-200ms	1.02	validation
burst5x20Hz_r50_dt-120ms	0.85	validation
burst5x20Hz_r50_dt-200ms	0.80	validation
10Hz_dt+10ms	0.90	validation
10Hz_dt-10ms	0.72	validation
20Hz_dt-10ms	0.65	validation
40Hz_dt-10ms	0.92	validation
50Hz_dt+10ms	1.22	validation
2Hz_5ms	1.28	validation
5Hz_5ms	0.98	validation
20Hz_5ms	0.88	validation
30Hz_5ms	1.40	validation
40Hz_5ms	1.45	validation
0.1Hz_dt0ms	1.02	validation
20Hz_dt+25ms	0.65	validation
20Hz_dt0ms	0.75	validation
20Hz_dt-25ms	0.68	validation
40Hz_dt0ms	1.05	validation
S07 pair mGluR bl.	0.92	validation
S07 pre_only	2.15	validation
S07 post_only	0.95	validation

Figure 3: A scatter plot showing the EPSP ratio (after/before) for various experimental conditions. The y-axis ranges from 0.7 to 1.4. The x-axis lists conditions: Lz 1ap_dt+10ms, Lz 3ap_200hz_dt+10ms@proximal, Lz 3ap_200hz_dt+10ms@distal, 50hz_dt+10ms@distal, Lz 3ap_200hz_dt-10ms@distal, 50hz_dt+10ms@proximal, 50hz_dt+10ms, Lz 3ap_200hz_dt+10ms@distal NMDAR bl., Lz nopost, and Lz 3ap_200hz_dt-500ms. Data points are red circles with error bars. A horizontal line is at 1.0. A legend indicates: 4N seeded, fitted (χ^2 24.1, n 9) (red circle), 4N seeded, validation (χ^2 0.0, n 0) (red circle with error bar), and data (mean $\pm$ SEM) (black error bar).

Figure 3 is a scatter plot showing the EPSP ratio (after/before) for various experimental conditions. The y-axis ranges from 0.6 to 2.0. The x-axis lists 15 conditions. The plot is divided into two regions: 'fitted' (left) and 'validation' (right), separated by a vertical dashed line. Data points are shown as green circles with error bars (SEM). Fitted curves are shown as solid lines. The legend indicates: 4N seeded, fitted (χ^2 107.0, n 15); 4N seeded, validation (χ^2 0.4, n 1); data (mean $\pm$ SEM).

Condition	EPSP ratio (after/before)	Region
Z 1ap_dt+10ms	~0.98	fitted
Z 1ap_dt-10ms	~0.75	fitted
Z train10_50hz_dt+4ms	~1.30	fitted
Z train10_50hz_dt-10ms	~0.85	fitted
Z pre_only	~1.00	fitted
Z 5ap_10hz_dt+10ms	~0.85	fitted
Z 5ap_20hz_dt+10ms	~1.30	fitted
Z train10_50hz_dt-10ms	~1.00	fitted
Z train10_50hz_dt-4ms	~0.88	fitted
Z train10_50hz_dt+5ms	~1.28	fitted
Z train4_50hz_post_only	~1.02	fitted
Z train8_50hz_dt+4ms	~0.88	fitted
Z train10_50hz_dt+4ms	~1.18	fitted
Z train10_50hz_dt-10ms mGluR bl.	~1.35	fitted
Egger 5ap_20hz_dt+10ms	~1.28	validation

Bar chart showing the contribution of different L5 inputs to the L2/3 population response. The y-axis is χ^2 contribution (z^2) from 0 to 160. The x-axis lists 48 input conditions. A legend indicates: L5→L5 (blue), L2/3→L5 (red), and L2/3→L2/3 (green). A dashed vertical line is at the 25th condition.

Input Condition	χ^2 contribution (z^2)	Category
Z dt-10ms mGluR bl.	28	L2/3→L2/3
rain10_50hz_dt+5ms	23	L2/3→L2/3
Z 1ap_dt+10ms	23	L2/3→L2/3
Lz 1ap_dt+10ms	14	L2/3→L2/3
10Hz_dt+10ms	10	L2/3→L2/3
Z 1ap_dt-10ms	10	L5→L5
50hz_dt-10ms	8	L2/3→L2/3
rain10_50hz_dt-10ms	7	L5→L5
1hz_dt+10ms@distal	4	L2/3→L2/3
5ap_20hz_dt+10ms	4	L2/3→L2/3
rain4_50hz_dt+10ms	4	L2/3→L2/3
dt+10ms_dt+4ms	3	L2/3→L2/3
S07 pair	3	L2/3→L2/3
rain10_50hz_dt+4ms	2	L2/3→L2/3
Z dt+4ms mGluR bl.	2	L2/3→L2/3
rain10_50hz_dt-4ms	2	L5→L5
Z dt-10ms@proximal	2	L2/3→L2/3
40hz_dt+10ms	2	L2/3→L2/3
S07 pair NO bl.	2	L2/3→L2/3
0.1hz_dt-10ms	2	L2/3→L2/3
Z 5ap_20hz_dt-10ms	2	L2/3→L2/3
5ap_10hz_dt+10ms	2	L2/3→L2/3
0.1hz_dt+10ms	2	L2/3→L2/3
Lz nopost	1	L2/3→L2/3
dt+10ms NMDAR bl.	0	L2/3→L2/3
ap_200hz_dt-500ms	0	L2/3→L2/3
rain10_50hz_dt-500ms	0	L2/3→L2/3
Z pre_only	0	L2/3→L2/3
rain8_50hz_post_only	0	L2/3→L2/3
1hz_dt+10ms@distal	0	L2/3→L2/3
20hz_dt+10ms	0	L2/3→L2/3
50hz_dt+10ms	0	L2/3→L2/3
ms@distal NMDAR bl.	0	L2/3→L2/3
Z dt-10ms mGluR bl.	0	L2/3→L2/3
40hz_dt0ms	155	L5→L5
20hz_dt0ms	30	L5→L5
20hz_dt-25ms	25	L5→L5
20hz_dt+25ms	10	L5→L5
20hz_dt-10ms	10	L5→L5
Z dt-10ms mGluR bl.	9	L5→L5
10hz_dt+10ms	8	L5→L5
20hz_r50_dt-200ms	7	L5→L5
10Hz_5ms	6	L5→L5
5Hz_5ms	4	L5→L5
dt+10ms mGluR bl.	3	L5→L5
dt-10ms NMDAR bl.	2	L5→L5
0.1hz_dt-25ms	2	L5→L5
10hz_dt-10ms	2	L5→L5
20hz_r50_dt-120ms	2	L5→L5
S07 pair mGluR bl.	2	L5→L5
50hz_dt+10ms	1	L5→L5
40hz_dt-10ms	1	L5→L5
Hz_-10ms mGluR bl.	1	L5→L5
0.1hz_dt0ms	1	L5→L5
S07 pre_only	1	L5→L5
5ap_20hz_dt+10ms	1	L5→L5
0.1hz_dt-200ms	1	L5→L5
0.1hz_dt-120ms	1	L5→L5
20Hz_5ms	1	L5→L5
40Hz_5ms	1	L5→L5
S07 post_only	1	L5→L5
30Hz_5ms	1	L5→L5
dt-10ms NMDAR bl.	1	L5→L5
2Hz_5ms	1	L5→L5